On the Nature of Polynomial Roots: A Chebyshev-Trigonometric Classification

Sawon Pratiher

Email ID: sawon1234@gmail.com

Abstract

This article provides a simple trigonometric method for determining how many roots of a polynomial equation are real and how many are complex, without solving the equation. For any depressed polynomial $P(t) = t^d + c_{d-2}t^{d-2} + \cdots + c_1 t + c_0 = 0$ of degree d with $c_{d-2} < 0$, the substitution $t = u\cos\theta$ with $u = 2\sqrt{-c_{d-2}/d}$ converts the root classification problem into the analysis of a trigonometric function $f(\theta) = \sum_{k=0}^{d-3} \alpha_k \cos^k\theta + \cos d\theta + \delta = 0$ on $[0, \pi]$, obtained by matching the polynomial to the Chebyshev identity $T_d(\cos\theta) = \cos d\theta$. The bounded range of the cosine function, combined with a universal exterior root lemma based on the positivity of $P^{(d-2)}$, yields a complete classification of the polynomial's roots for every degree. The derivation is computationally light and conceptually natural, unifying and extending the quartic and quintic analyses to all degrees $d \geq 4$.

1 Introduction

How many roots of a given polynomial are real? For the quadratic, the discriminant $b^2 - 4ac$ answers the question instantly. Loh [20] showed that even the derivation of the quadratic formula can be made simple enough for first time Algebra learners. For higher degree polynomials the discriminant becomes unwieldy: it is a polynomial of degree 6 in the coefficients for the quartic [25, 26], degree 8 for the quintic, and degree $d(d-1)$ in general.

This article introduces a uniform alternative. We show that a single trigonometric substitution $t = u\cos\theta$ for an explicit constant u converts the root classification problem for *any* degree d polynomial into the analysis of a trigonometric function on $[0, \pi]$. The derivation uses nothing beyond the Chebyshev identity $T_d(\cos\theta) = \cos d\theta$, the intermediate value theorem, and a short computation showing that a certain high order derivative is positive outside a bounded interval.

The idea has a simple geometric interpretation: the map $\theta \mapsto u\cos\theta$ wraps the interval $[0, \pi]$ onto $[-u, u]$, and the Chebyshev identity absorbs the leading and subleading terms of the depressed polynomial, leaving a "residual" trigonometric equation whose zeros correspond to roots of the polynomial. Roots outside $[-u, u]$ are separately controlled by a monotonicity argument.

The method was developed for the quartic ($d = 4$) [34] and the quintic ($d = 5$) [35] individually in earlier work. In this article, we state and prove the general framework for all $d \geq 4$, with the sextic ($d = 6$) as the primary detailed example. The reader is encouraged to notice how the same algebraic skeleton substitution, Chebyshev matching, bijection, boundary sign analysis applies uniformly across all degrees, with only the number of residual terms and the order of the exterior root derivative changing. Throughout, all polynomials have real coefficients, so complex roots occur

in conjugate pairs. A polynomial of degree d therefore has $d, d-2, d-4, \ldots$ or $d \bmod 2$ real roots. (The probabilistic question how many real roots does a *random* polynomial have? has a rich history of its own, from Kac [16] to Tao and Vu, but is not our concern here.)

2 The general framework

2.1 Depressed polynomials

Every monic polynomial of degree d can be reduced to *depressed* form with no t^{d-1} term by the Tschirnhaus substitution $t = z - a_{d-1}/d$, where a_{d-1} is the coefficient of z^{d-1}. The result is

$$P(t) = t^d + c_{d-2}\, t^{d-2} + c_{d-3}\, t^{d-3} + \cdots + c_1\, t + c_0 = 0. \tag{1}$$

By Vieta's formulas, the roots satisfy $t_1 + t_2 + \cdots + t_d = 0$, so the centroid of the roots sits at the origin.

2.2 The Chebyshev identity

For each positive integer d, the Chebyshev polynomial of the first kind $T_d(x)$ satisfies $T_d(\cos\theta) = \cos d\theta$ for all θ. Its leading and subleading terms are (see, e.g., [21]):

$$T_d(x) = 2^{d-1}\, x^d - d \cdot 2^{d-3}\, x^{d-2} + \text{lower order terms}. \tag{2}$$

The identity $T_d(\cos\theta) = \cos d\theta$ is the engine of our method.

2.3 The substitution

Assume $c_{d-2} < 0$ and set

$$\boxed{u = 2\sqrt{\frac{-c_{d-2}}{d}}\,.} \tag{3}$$

Note $u^2 = -4c_{d-2}/d > 0$. Substituting $t = u\cos\theta$ into (1) and dividing by $u^d/2^{d-1}$:

$$2^{d-1}\cos^d\theta + \frac{2^{d-1}c_{d-2}}{u^2}\cos^{d-2}\theta + \sum_{k=0}^{d-3} \frac{2^{d-1}c_k}{u^{d-k}}\cos^k\theta = 0. \tag{4}$$

The choice of u ensures

$$\frac{2^{d-1}c_{d-2}}{u^2} = \frac{2^{d-1}c_{d-2}}{-4c_{d-2}/d} = -d \cdot 2^{d-3},$$

which is exactly the subleading coefficient of $T_d(x)$. Therefore the first two terms of (4) match those of $T_d(\cos\theta)$. Subtracting the Chebyshev identity, we obtain the **reduced trigonometric equation**:

$$\boxed{f(\theta) = \sum_{k=0}^{d-3} \alpha_k \cos^k\theta + \cos d\theta + \delta = 0,} \tag{5}$$

where the **trigonometric parameters** $\alpha_0, \ldots, \alpha_{d-3}$ and δ are explicit linear functions of the polynomial coefficients $c_0, \ldots, c_{d-3}$:

$$\alpha_k = \frac{2^{d-1} c_k}{u^{d-k}} - [T_d]_k \qquad (k = 0, 1, \ldots, d-3), \tag{6}$$

with $\delta = \alpha_0$ and $[T_d]_k$ denoting the coefficient of x^k in $T_d(x)$. The constant δ absorbs the constant terms from both the polynomial and the Chebyshev identity.

Remark 1. The reduced equation has *exactly* $d-2$ free parameters (corresponding to the $d-2$ coefficients $c_0, \ldots, c_{d-3}$ of the depressed polynomial). The two "absorbed" coefficients the leading $c_d = 1$ and the subleading c_{d-2} are consumed by matching the Chebyshev identity. The number of residual cosine terms grows with the degree:

d	Residual terms in $f(\theta)$
4	$\cos\theta$
5	$\cos^2\theta,\ \cos\theta$
6	$\cos^3\theta,\ \cos^2\theta,\ \cos\theta$
7	$\cos^4\theta,\ \cos^3\theta,\ \cos^2\theta,\ \cos\theta$
d	$\cos^{d-3}\theta,\ \ldots,\ \cos\theta$

2.4 The fundamental identity

The following identity is the bridge between the polynomial and trigonometric worlds.

Proposition 2. *For all* $\theta \in \mathbb{R}$,

$$f(\theta) = \frac{2^{d-1}}{u^d} P(u\cos\theta). \tag{7}$$

Proof. The left side of (4) equals $2^{d-1}P(u\cos\theta)/u^d$ by construction. Rewriting the first two terms via the Chebyshev identity gives $\cos d\theta$ plus the remaining Chebyshev terms; the residual is precisely the left side of (5). □

Corollary 3. *Since* $\cos\colon [0,\pi] \to [-1,1]$ *is a continuous, strictly decreasing bijection, the zeros of* f *in* $[0,\pi]$ *are in bijection with the roots of* P *in* $[-u,u]$.

2.5 Boundary values

Evaluating (7) at the endpoints:

$$f(0) = \frac{2^{d-1}P(u)}{u^d}, \qquad f(\pi) = \frac{2^{d-1}P(-u)}{u^d}. \tag{8}$$

Since $u^d > 0$ and $2^{d-1} > 0$, the sign of $f(0)$ equals the sign of $P(u)$, and the sign of $f(\pi)$ equals the sign of $P(-u)$.

3 The exterior root lemma

The substitution captures roots in $[-u, u]$, but the polynomial may have real roots outside this interval. The following lemma, valid for *all* degrees $d \geq 2$, controls the number of such roots.

Lemma 4 (Universal exterior positivity)**.** *For any depressed polynomial $P(t) = t^d + c_{d-2}t^{d-2} + \cdots$ with $c_{d-2} < 0$ and $u = 2\sqrt{-c_{d-2}/d}$,*

$$P^{(d-2)}(t) > 0 \qquad \textit{for all } |t| > u. \tag{9}$$

Proof. Since only the terms t^d and $c_{d-2}t^{d-2}$ survive $d-2$ differentiations:

$$P^{(d-2)}(t) = \frac{d!}{2}\, t^2 + (d-2)!\, c_{d-2}\,.$$

For $|t| > u$, we have $t^2 > u^2 = -4c_{d-2}/d$, so

$$\begin{aligned} P^{(d-2)}(t) &> \frac{d!}{2} \cdot \frac{-4c_{d-2}}{d} + (d-2)!\, c_{d-2} \\ &= c_{d-2}\big[-2(d-1)! + (d-2)!\big] \\ &= c_{d-2}\,(d-2)!\,(3-2d). \end{aligned}$$

Since $c_{d-2} < 0$ and $3 - 2d < 0$ for $d \geq 2$, the product is positive. □

The power of this lemma is that it applies uniformly to every degree. Its consequences cascade through the derivatives:

Corollary 5. *On each of the half lines (u, ∞) and $(-\infty, -u)$:*

(a) *$P^{(d-2)}$ is positive, so $P^{(d-3)}$ is strictly monotone.*

(b) *$P^{(d-3)}$ has at most one zero, so $P^{(d-4)}$ has at most one critical point and hence at most two zeros.*

(c) *Continuing inductively: $P^{(k)}$ has at most $d-2-k$ zeros on each half line, for $k = 0, 1, \ldots, d-2$.*

(d) *In particular, P itself has at most $d-2$ zeros on each half line.*

In practice, for generic coefficient values, each half line contributes at most one root.

Proof. Part (a) follows directly from Lemma 4. For (b)-(d), apply Rolle's theorem inductively: between any two consecutive zeros of $P^{(k)}$ lies a zero of $P^{(k+1)}$, so the zero count decreases by at most one at each differentiation step. □

3.1 Parity and the intermediate value theorem

The number of roots on each half line has a definite parity, determined by the boundary value of P (or equivalently of f) and the end behavior:

Proposition 6 (Exterior root detection)**.** *Let $d \geq 2$.*

(i) *On (u, ∞): since $P(t) \to +\infty$ as $t \to +\infty$, the number of roots is odd if $P(u) < 0$ (i.e., $f(0) < 0$) and even if $P(u) \geq 0$ (i.e., $f(0) \geq 0$).*

(ii) *On $(-\infty, -u)$: since $P(t) \to (-1)^d \cdot \infty$ as $t \to -\infty$, the parity depends on both $P(-u)$ and the degree. Specifically:*

- *If d is even: the number of roots is odd if $P(-u) < 0$ (i.e., $f(\pi) < 0$) and even if $P(-u) \geq 0$.*
- *If d is odd: the number of roots is odd if $P(-u) > 0$ (i.e., $f(\pi) > 0$) and even if $P(-u) \leq 0$.*

In the generic case (at most one root per half line), the exterior root count is:

$$N_{\text{ext}}^{+} = \mathbf{1}_{f(0)<0}\,, \tag{10}$$

and for the left half line:

$$N_{\text{ext}}^{-} = \begin{cases} \mathbf{1}_{f(\pi)<0} & \text{if } d \text{ is even}, \\ \mathbf{1}_{f(\pi)>0} & \text{if } d \text{ is odd}. \end{cases} \tag{11}$$

4 Critical point analysis

To count the zeros of f on $[0, \pi]$, we study its critical points. Differentiating (5) and factoring out $\sin\theta > 0$ on $(0, \pi)$:

Proposition 7. *The interior critical points of f in $(0, \pi)$ are the solutions of a polynomial equation of degree $d-1$ in $x = \cos\theta$. Consequently, f has at most $d-1$ interior critical points, and therefore at most d zeros in $[0, \pi]$ consistent with P having at most d real roots.*

Proof. The derivative is

$$f'(\theta) = -\sum_{k=1}^{d-3} k\,\alpha_k \cos^{k-1}\theta\,\sin\theta - d\sin d\theta.$$

Using $\sin d\theta = \sin\theta \cdot U_{d-1}(\cos\theta)$, where U_{d-1} is the Chebyshev polynomial of the second kind of degree $d-1$, and dividing by $-\sin\theta$ (positive on $(0, \pi)$), the critical points satisfy a polynomial equation of degree $d-1$ in $x = \cos\theta \in (-1, 1)$. □

5 Classification theorem

We now state the general result. Let N_{int} denote the number of zeros of f in $[0, \pi]$ and N_{ext} the number of roots of P outside $[-u, u]$.

Theorem 8 (General root classification)**.** *Let $P(t) = t^d + c_{d-2}t^{d-2} + \cdots + c_0$ be a depressed polynomial of degree d with real coefficients and $c_{d-2} < 0$. Set $u = 2\sqrt{-c_{d-2}/d}$ and define $f(\theta)$ by (5). Then the total number of real roots of P is*

$$N_{\text{real}} = N_{\text{int}} + N_{\text{ext}}\,, \tag{12}$$

where N_{int} is the number of zeros of f in $[0, \pi]$ and $N_{\text{ext}} = N_{\text{ext}}^{+} + N_{\text{ext}}^{-}$ is determined by the signs of $f(0)$ and $f(\pi)$ via Proposition 6. The possible values of N_{real} are $d, d-2, d-4, \ldots$, down to 0 (if d is even) or 1 (if d is odd).

Proof. Corollary 3 gives the bijection between interior zeros of f and roots of P in $[-u, u]$. The exterior roots are counted by Proposition 6, with the bound from Corollary 5. The parity constraint follows from complex roots occurring in conjugate pairs. □

A useful sufficient condition for "few" real roots is:

Corollary 9. *If d is even and $f(\theta) > 0$ for all $\theta \in [0, \pi]$, then P has no real roots. A sufficient condition is*

$$\delta > \sum_{k=1}^{d-3} |\alpha_k| + 1, \tag{13}$$

since $|\alpha_k \cos^k\theta + \cos d\theta| \le \sum |\alpha_k| + 1$.

5.1 Practical decision procedure

For any depressed polynomial of degree d with $c_{d-2} < 0$:

1. Compute $u = 2\sqrt{-c_{d-2}/d}$ and the trigonometric parameters from (6).
2. Evaluate $f(0)$ and $f(\pi)$ to determine N_{ext}.
3. Find the critical points of f in $(0, \pi)$ by solving the degree $(d-1)$ polynomial in $x = \cos\theta$.
4. Evaluate f at the critical points and at $\theta = 0, \pi$. Count sign changes to determine N_{int}.
5. Conclude: $N_{\text{real}} = N_{\text{int}} + N_{\text{ext}}$.

6 Exhaustive case by case analysis

We now work through every degree from $d = 3$ to $d = 8$ in complete detail, then summarize the general pattern. For each degree we give: the Chebyshev identity, the substitution, the reduced trigonometric equation, the exterior root derivative, the critical point equation, the classification, and a representative example.

6.1 The cubic ($d = 3$)

The depressed cubic is $P(t) = t^3 + pt + q = 0$ (see [8] for a historical account). The Chebyshev identity is $T_3(\cos\theta) = 4\cos^3\theta - 3\cos\theta = \cos 3\theta$. Setting $u = 2\sqrt{-p/3}$ (requires $p < 0$) and substituting $t = u\cos\theta$:

$$f(\theta) = \cos 3\theta + \gamma = 0, \qquad \gamma = \frac{4q}{u^3}\,. \tag{14}$$

There are *no* residual cosine terms the cubic is the unique degree for which the Chebyshev identity absorbs everything except a constant. This is precisely Vieta's trigonometric solution: the roots are $t_k = u\cos(\theta_k)$, where $\cos 3\theta_k = -\gamma$.

Exterior analysis. $P'(t) = 3t^2 + p > 0$ for $|t| > u$, so P is strictly monotone on each half line, giving at most one exterior root per side. Since $d = 3$ is odd: $N^+_{\text{ext}} = \mathbf{1}_{f(0)<0}$, $N^-_{\text{ext}} = \mathbf{1}_{f(\pi)>0}$.

Critical points. The critical point equation is the quadratic $U_2(x) = 4x^2 - 1 = 0$ in $x = \cos\theta$, giving at most two interior critical points and hence at most three zeros of f in $[0, \pi]$.

Classification. $N_{\text{real}} \in \{1, 3\}$. All three roots are real iff $|\gamma| \le 1$, i.e., $4q^2 \le u^6$, i.e., $4q^2 + 4p^3/27 \le 0$ the classical cubic discriminant condition $\Delta = -4p^3 - 27q^2 \ge 0$.

Example 10. $P(t) = t^3 - 3t + 1 = 0$. Here $p = -3$, $q = 1$, $u = 2$, $\gamma = 1/2$. Since $|\gamma| = 1/2 < 1$, all three roots are real. The equation $\cos 3\theta = -1/2$ has solutions $\theta = 2\pi/9, 4\pi/9, 8\pi/9$ in $[0, \pi]$, giving $t \approx 1.532, 0.347, -1.879$.

6.2 The quartic ($d = 4$)

The depressed quartic is $P(t) = t^4 + mt^2 + pt + q = 0$ (see [34] for a dedicated treatment and [17] for broader context). The Chebyshev identity is $T_4(\cos\theta) = 8\cos^4\theta - 8\cos^2\theta + 1 = \cos 4\theta$. Setting $u = \sqrt{-m}$ (requires $m < 0$) and substituting $t = u\cos\theta$:

$$f(\theta) = a\cos\theta + \cos 4\theta + b = 0, \tag{15}$$

where

$$a = \frac{8p}{(-m)^{3/2}}\,, \qquad b = \frac{8q}{m^2} - 1\,. \tag{16}$$

There is exactly **one** residual term ($\cos\theta$).

Exterior analysis. $P''(t) = 12t^2 + 2m > 0$ for $|t| > u$, so P is strictly convex on each half line, giving at most one exterior root per side. Since $d = 4$ is even: $N^{\pm}_{\text{ext}} = \mathbf{1}_{f(0,\pi)<0}$.

Critical points. The critical point equation is the cubic $h(x) = x(2x^2 - 1) = -a/16$ in $x = \cos\theta$, with at most three interior critical points and hence at most four zeros of f in $[0, \pi]$.

Classification. $N_{\text{real}} \in \{0, 2, 4\}$. All four roots are complex iff $f > 0$ on $[0, \pi]$; sufficient: $b > |a| + 1$.

Example 11. $P(t) = t^4 - 25t^2 - 60t - 36 = 0$. Here $m = -25$, $p = -60$, $q = -36$, $u = 5$. The parameters are $a = -3.84$, $b = -1.4608$. The boundary values $f(0) = -4.30 < 0$ (one exterior root in $(5, \infty)$) and $f(\pi) = 3.38 > 0$ (no left exterior root). Numerical analysis gives $N_{\text{int}} = 3$, so $N_{\text{real}} = 4$. The roots are $t = -1, -2, -3, 6$.

6.3 The quintic ($d = 5$)

The depressed quintic is $P(t) = t^5 + mt^3 + nt^2 + pt + q = 0$ (see [35] for a dedicated treatment). The Chebyshev identity is $T_5(\cos\theta) = 16\cos^5\theta - 20\cos^3\theta + 5\cos\theta = \cos 5\theta$. Setting $u = 2\sqrt{-m/5}$

(requires $m < 0$) and substituting $t = u\cos\theta$:

$$f(\theta) = \alpha\cos^2\theta + \beta\cos\theta + \cos 5\theta + \gamma = 0, \tag{17}$$

where

$$\alpha = \frac{16n}{u^3}\,, \qquad \beta = \frac{16p}{u^4} - 5\,, \qquad \gamma = \frac{16q}{u^5}\,. \tag{18}$$

There are **two** residual terms ($\cos^2\theta$ and $\cos\theta$). When $n = 0$, the $\cos^2\theta$ term vanishes and the equation has the same form as the quartic's.

Exterior analysis. $P'''(t) = 60t^2 + 6m > 0$ for $|t| > u$. Since $d = 5$ is odd: $N_{\text{ext}}^{+} = \mathbf{1}_{f(0)<0}$, $N_{\text{ext}}^{-} = \mathbf{1}_{f(\pi)>0}$ (note the asymmetry).

Critical points. The critical point equation is the quartic $80x^4 - 60x^2 + 2\alpha x + (\beta + 5) = 0$ in $x = \cos\theta$, with at most four interior critical points and hence at most five zeros in $[0, \pi]$.

Classification. $N_{\text{real}} \in \{1, 3, 5\}$. The guaranteed existence of at least one real root follows from odd degree.

A perfect example: $P(t) = t^5 - 5t^3 + 5t = 0$ gives $\alpha = \beta = \gamma = 0$ and $f(\theta) = \cos 5\theta = 0$, with five zeros at $\theta = k\pi/10$ ($k = 1, 3, 5, 7, 9$) and roots $t = 0, \pm\sqrt{(5 \pm \sqrt{5})/2}$.

Example 12. $P(t) = t^5 - 5t^3 + t - 5 = 0$. Here $m = -5$, $n = 0$, $p = 1$, $q = -5$, $u = 2$. The parameters are $\alpha = 0$, $\beta = -4$, $\gamma = -5/2$. The boundary values $f(0) = -11/2 < 0$ and $f(\pi) = 1/2 > 0$ give $N_{\text{ext}} = 2$. One interior zero gives $N_{\text{int}} = 1$ and $N_{\text{real}} = 3$. The roots are $t \approx -2.043, -1.205, 2.286$.

6.4 The sextic ($d = 6$)

The depressed sextic is $P(t) = t^6 + mt^4 + nt^3 + pt^2 + qt + r = 0$. The Chebyshev identity is $T_6(\cos\theta) = 32\cos^6\theta - 48\cos^4\theta + 18\cos^2\theta - 1 = \cos 6\theta$. Setting $u = \sqrt{-2m/3}$ (requires $m < 0$) and substituting $t = u\cos\theta$:

$$f(\theta) = \alpha\cos^3\theta + \beta\cos^2\theta + \gamma\cos\theta + \cos 6\theta + \delta = 0, \tag{19}$$

where

$$\alpha = \frac{32n}{u^3}\,, \qquad \beta = \frac{32p}{u^4} - 18\,, \qquad \gamma = \frac{32q}{u^5}\,, \qquad \delta = \frac{32r}{u^6} + 1\,. \tag{20}$$

There are **three** residual terms. When $n = q = 0$ (no odd power terms), we have $\alpha = \gamma = 0$ and f is even in θ, reflecting the symmetry $P(t) = P(-t)$.

Exterior analysis. $P^{(4)}(t) = 360t^2 + 24m > 0$ for $|t| > u$. Since $d = 6$ is even: $N_{\text{ext}}^{\pm} = \mathbf{1}_{f(0,\pi)<0}$.

Critical points. The critical point equation is the quintic $192x^5 - 192x^3 + 36x + 3\alpha x^2 + 2\beta x + \gamma = 0$ in $x = \cos\theta$, with at most five interior critical points and at most six zeros in $[0, \pi]$.

Classification. $N_{\text{real}} \in \{0, 2, 4, 6\}$. All roots complex iff $f > 0$ on $[0, \pi]$; sufficient: $\delta > |\alpha| + |\beta| + |\gamma| + 1$.

A perfect example: $(t^2 - 1)(t^2 - 2)(t^2 - 3) = t^6 - 6t^4 + 11t^2 - 6$ gives $u = 2$, $\alpha = 0$, $\beta = 4$, $\gamma = 0$, $\delta = -2$, and $f(\theta) = 4\cos^2\theta + \cos 6\theta - 2 = 0$ has six zeros in $[0, \pi]$. All six roots $t = \pm 1, \pm\sqrt{2}, \pm\sqrt{3}$ lie in $[-2, 2]$.

Example 13. $P(t) = t^6 - 6t^4 + 5t^2 + 1 = 0$. Here $u = 2$, $\alpha = \gamma = 0$, $\beta = -8$, $\delta = 1.5$. The boundary values $f(0) = f(\pi) = -5.5 < 0$ give $N_{\text{ext}} = 2$. Numerical analysis gives $N_{\text{int}} = 2$ and $N_{\text{real}} = 4$. The roots are $t \approx \pm 1.103, \pm 2.225$.

Example 14. $P(t) = t^6 - 6t^4 + 10t^2 + 1 = 0$. Here $u = 2$, $\alpha = \gamma = 0$, $\beta = 2$, $\delta = 1.5$. The boundary values $f(0) = f(\pi) = 4.5 > 0$, and $f > 0$ on all of $[0, \pi]$ (minimum ≈ 0.5 at $\theta = \pi/2$), so $N_{\text{real}} = 0$: all six roots are complex.

Example 15. $P(t) = t^6 - 6t^4 + t^3 + 9t^2 - 3t - 1 = 0$. Here $u = 2$, $\alpha = 4$, $\beta = 0$, $\gamma = -3$, $\delta = 0.5$. Since $4\cos^3\theta - 3\cos\theta = \cos 3\theta$ by the triple angle formula, the reduced equation simplifies to $f(\theta) = \cos 3\theta + \cos 6\theta + 0.5 = 0$. Boundary values $f(0) = 2.5 > 0$, $f(\pi) = 0.5 > 0$ give $N_{\text{ext}} = 0$. All six real roots $t \approx -1.956, -1.618, -0.209, 0.618, 1.338, 1.827$ lie in $[-2, 2]$.

6.5 The septic ($d = 7$)

The depressed septic is $P(t) = t^7 + mt^5 + nt^4 + pt^3 + qt^2 + rt + s = 0$. The Chebyshev identity is $T_7(\cos\theta) = 64\cos^7\theta - 112\cos^5\theta + 56\cos^3\theta - 7\cos\theta = \cos 7\theta$. Setting $u = 2\sqrt{-m/7}$ (requires $m < 0$) and substituting $t = u\cos\theta$:

$$f(\theta) = \alpha_4\cos^4\theta + \alpha_3\cos^3\theta + \alpha_2\cos^2\theta + \alpha_1\cos\theta + \cos 7\theta + \delta = 0, \tag{21}$$

where $\alpha_4 = 64n/u^4$, $\alpha_3 = 64p/u^5 - 56$, $\alpha_2 = 64q/u^6$, $\alpha_1 = 64r/u^7 + 7$, and $\delta = 64s/u^8$. There are **four** residual terms.

Exterior analysis. $P^{(5)}(t) = 2520t^2 + 120m > 0$ for $|t| > u$. Since $d = 7$ is odd: $N_{\text{ext}}^+ = \mathbf{1}_{f(0)<0}$, $N_{\text{ext}}^- = \mathbf{1}_{f(\pi)>0}$.

Critical points. The critical point equation is of degree 6 in $x = \cos\theta$ (from $U_6(x) = 64x^6 - 80x^4 + 24x^2 - 1$ plus residual derivatives), giving at most six interior critical points and at most seven zeros in $[0, \pi]$.

Classification. $N_{\text{real}} \in \{1, 3, 5, 7\}$.

Example 16. $P(t) = t^7 - 7t^5 + 10t^3 - 5t = 0$. Here $m = -7$, $u = 2$, and the nonzero terms are $n = q = s = 0$, $p = 10$, $r = -5$. We compute $\alpha_4 = 0$, $\alpha_3 = 64 \cdot 10/32 - 56 = -36$, $\alpha_2 = 0$, $\alpha_1 = 64 \cdot (-5)/128 + 7 = 4.5$, $\delta = 0$. The boundary values $f(0) = 0 - 36 + 0 + 4.5 + 1 + 0 = -30.5$ and $f(\pi) = 0 - 36 + 0 - 4.5 - 1 + 0 = -41.5$ indicate exterior roots. Numerical computation confirms $N_{\text{real}} = 3$: the roots are $t \approx -2.299, 0, 2.299$.

6.6 The octic ($d = 8$)

The depressed octic is $P(t) = t^8 + mt^6 + nt^5 + pt^4 + qt^3 + rt^2 + st + w = 0$. The Chebyshev identity is $T_8(\cos\theta) = 128\cos^8\theta - 256\cos^6\theta + 160\cos^4\theta - 32\cos^2\theta + 1 = \cos 8\theta$. Setting $u = \sqrt{-m/2}$ (requires $m < 0$) and substituting $t = u\cos\theta$:

$$f(\theta) = \sum_{k=1}^{5} \alpha_k \cos^k\theta + \cos 8\theta + \delta = 0, \tag{22}$$

where $\alpha_5 = 128n/u^5$, $\alpha_4 = 128p/u^6 - 160$, $\alpha_3 = 128q/u^7$, $\alpha_2 = 128r/u^8 + 32$, $\alpha_1 = 128s/u^9$, and $\delta = 128w/u^{10} - 1$. There are **five** residual terms.

Exterior analysis. $P^{(6)}(t) = 20160t^2 + 720m > 0$ for $|t| > u$. Since $d = 8$ is even: $N_{\text{ext}}^{\pm} = \mathbf{1}_{f(0,\pi)<0}$.

Critical points. The critical point equation is of degree 7 in $x = \cos\theta$ (from $U_7(x) = 128x^7 - 192x^5 + 80x^3 - 8x$ plus residual derivatives), giving at most seven interior critical points and at most eight zeros in $[0, \pi]$.

Classification. $N_{\text{real}} \in \{0, 2, 4, 6, 8\}$.

A perfect example: $(t^2 - \frac{1}{2})(t^2 - \frac{3}{2})(t^2 - \frac{5}{2})(t^2 - \frac{7}{2}) = t^8 - 8t^6 + 21.5t^4 - 22t^2 + 6.5625$ gives $u = 2$ and eight real roots $t = \pm\sqrt{k + 1/2}$ for $k = 0, 1, 2, 3$, all in $[-2, 2]$.

Example 17. $P(t) = t^8 - 8t^6 + 18t^4 + 1 = 0$. Here $m = -8$, $u = 2$, and with only even powers present, all odd index parameters vanish. We get $\alpha_4 = 128 \cdot 18/64 - 160 = -124$, $\alpha_2 = 32$, $\delta = 128/1024 - 1 = -0.875$. Since $f(0)$ and $f(\pi)$ are both positive (by symmetry) and $f > 0$ on $[0, \pi]$, we have $N_{\text{real}} = 0$: all eight roots are complex.

6.7 Synthesis: the pattern across all degrees

The following table collects the key features for degrees 3 through 10 and the general degree d.

d	u^2	Residual terms	Crit. pt degree	Exterior deriv.	Real root counts
3	$-4m/3$	0	2	$P' > 0$	$1, 3$
4	$-m$	1	3	$P'' > 0$	$0, 2, 4$
5	$-4m/5$	2	4	$P''' > 0$	$1, 3, 5$
6	$-2m/3$	3	5	$P^{(4)} > 0$	$0, 2, 4, 6$
7	$-4m/7$	4	6	$P^{(5)} > 0$	$1, 3, 5, 7$
8	$-m/2$	5	7	$P^{(6)} > 0$	$0, 2, \ldots, 8$
9	$-4m/9$	6	8	$P^{(7)} > 0$	$1, 3, \ldots, 9$
10	$-2m/5$	7	9	$P^{(8)} > 0$	$0, 2, \ldots, 10$
d	$\frac{-4c_{d-2}}{d}$	$d - 3$	$d - 1$	$P^{(d-2)} > 0$	$d \bmod 2, \ldots, d$

Several patterns are visible. The number of residual terms grows as $d - 3$, starting from zero for the cubic (the unique "clean" case). The critical point equation is always of degree $d - 1$ in $\cos\theta$. The exterior root derivative is always $P^{(d-2)}(t) = \frac{d!}{2}t^2 + (d-2)!\, c_{d-2}$, and the proof of its positivity outside $[-u, u]$ is identical for every d (Lemma 4). The set of possible real root counts is $\{d \bmod 2,\ d \bmod 2 + 2,\ \ldots,\ d\}$, reflecting the conjugate pairing constraint.

7 The case $c_{d-2} \geq 0$

When $c_{d-2} \geq 0$, the substitution $t = u\cos\theta$ with real u is not available. A hyperbolic substitution $t = v\cosh\phi$ with $v = 2\sqrt{c_{d-2}/d}$ could match the subleading coefficient via the analogous identity $T_d(\cosh\phi) = \cosh d\phi$, but since $\cosh$ is unbounded and monotone on $[0, \infty)$, the correspondence between zeros and roots is less clean. When $c_{d-2} = 0$, the Chebyshev identity cannot absorb the subleading term (since there is none), and the method does not apply directly.

For even degree polynomials with $c_{d-2} > 0$, one can show that $P^{(d-2)}(t) = \frac{d!}{2}t^2 + (d-2)!c_{d-2} > 0$ for *all* t, so P is "almost convex" in a strong sense, which often limits the number of real roots. We leave the systematic development of the $c_{d-2} \geq 0$ case to future work.

8 Relation to prior work

The problem of determining how many roots of a polynomial are real without computing the roots themselves has a long history stretching back several centuries. This section surveys the principal existing methods and clarifies the novelty of the trigonometric approach developed in this article.

8.1 The discriminant

The oldest and most direct tool is the polynomial discriminant. For the quadratic $ax^2 + bx + c = 0$, the discriminant $\Delta = b^2 - 4ac$ is a single expression whose sign immediately determines the nature of the roots. For the cubic, the discriminant $\Delta_3 = b^2c^2 - 4ac^3 - 4b^3d - 27a^2d^2 + 18abcd$ remains manageable [15]. For the quartic, the discriminant is a polynomial of degree 6 in the coefficients, with 16 terms [25]. For the quintic, it has 59 terms; for the sextic, 246 terms (see [11]). In general, the discriminant of a degree d polynomial is of degree $d(d-1)$ in its coefficients. The discriminant's role in the geometry of polynomial roots is intimately connected to the resultant, as elucidated in the influential monograph of Gelfand, Kapranov, and Zelevinsky [11]; Sylvester's classical work [32] on syzygetic relations provided the foundational algebraic framework; Marden [22] gives a comprehensive treatment of the geometry of polynomial zeros. Beyond degree 4, the discriminant alone does not determine the number of real roots: for the quintic, the sign of the discriminant distinguishes between one real root and five, but additional subdiscriminants are needed to distinguish one from three [18]. The computational and conceptual burden grows rapidly with the degree.

8.2 Descartes' rule of signs

Descartes' rule of signs (1637) provides an upper bound on the number of positive real roots: the number of sign changes in the coefficient sequence of $P(x)$ exceeds the number of positive real roots by a nonnegative even integer [7]. Applying the rule to $P(-x)$ gives a bound for negative roots. The rule is elegant and effortless to apply, but it gives only a bound, not an exact count. For instance, the polynomial $x^4 + 2x^2 - x - 1$ has one sign change, guaranteeing exactly one positive root, but for polynomials with many sign changes the gap between the bound and the true count can be large.

8.3 Sturm's theorem

Sturm's theorem (1829) provides an *exact* count of the distinct real roots of a polynomial in any interval [30]. Hermite [13] later reformulated the problem using quadratic forms, connecting the signature of a Hankel matrix to the number of real roots. The method constructs a sequence of polynomials (the Sturm chain) via a variant of the Euclidean algorithm applied to P and its derivative P', and the number of real roots in an interval $(a, b]$ equals the difference in the number of sign changes of the Sturm chain evaluated at a and at b. Sturm's theorem is complete and exact, but the Sturm chain requires d polynomial divisions, resulting in a sequence of polynomials whose coefficients can swell exponentially. For polynomials of degree exceeding about 20, modern implementations based on Descartes' rule with interval subdivision are substantially more efficient [2].

8.4 The Budan-Fourier theorem

The Budan-Fourier theorem (Budan 1807, Fourier 1820) refines Descartes' rule to any half open interval $(a, b]$: the number of real roots in the interval is bounded above by the difference in the number of sign changes of the sequence $P(x), P'(x), \ldots, P^{(d)}(x)$ evaluated at a and at b, with the discrepancy again being a nonnegative even integer. This is more informative than Descartes' rule but still provides only a bound, not an exact count (except in special cases).

8.5 Vieta's trigonometric solution of the cubic

The substitution $t = u\cos\theta$ has been used to *solve* the depressed cubic $t^3 + pt + q = 0$ since at least François Viète's work in the late 16th century [23, 36]. With $u = 2\sqrt{-p/3}$, the cubic becomes equivalent to $\cos 3\theta = -q/(2(-p/3)^{3/2})$, and the three roots are $t_k = 2\sqrt{-p/3}\cos(\theta_0/3 + 2k\pi/3)$ for $k = 0, 1, 2$. This is Vieta's celebrated trigonometric formula, valid in the *casus irreducibilis* when all three roots are real. Holmes [14] extended this to the one real root case using hyperbolic cosines. Crucially, Vieta's method *solves* the cubic (finding the roots explicitly); our method *classifies* the roots (counting how many are real) without finding them.

8.6 Chebyshev polynomials in numerical rootfinding

In the numerical analysis literature, Chebyshev polynomials are used extensively for polynomial rootfinding on bounded intervals. The Chebyshev-Frobenius companion matrix method, independently derived by Day and Romero and by Stetter, computes all roots of a polynomial expressed in Chebyshev form as eigenvalues of a structured matrix [5]. Boyd developed subdivision algorithms that partition the interval $[-1, 1]$ and approximate the polynomial locally by low degree Chebyshev interpolants, achieving efficient rootfinding for polynomials of high degree [5]. Trefethen's Chebfun system [33] implements these ideas in a widely used software package. However, all of this work is concerned with *computing* the roots (numerically), not with *classifying* their nature (real versus complex) without finding them.

8.7 What is new in this article

Against this background, the novelty of the present work can be stated precisely. The idea of using the substitution $t = u\cos\theta$ to *classify* rather than solve polynomial roots via the Chebyshev identity $T_d(\cos\theta) = \cos d\theta$ appears, to the best of the author's knowledge, to be new. Specifically:

1. **The Chebyshev matching reduction.** The systematic use of $T_d(\cos\theta) = \cos d\theta$ to absorb the two leading terms of a depressed polynomial and produce a reduced trigonometric equation $f(\theta)$ whose zeros on $[0, \pi]$ biject with real roots in $[-u, u]$ has not been previously exploited for root classification. Vieta's trigonometric method for the cubic uses the $d = 3$ case to *solve* the equation; we use the $d \geq 4$ cases to *classify* roots.

2. **The universal exterior root lemma.** The observation that $P^{(d-2)}(t) > 0$ for $|t| > u$ (Lemma 4), valid for all degrees $d \geq 2$ and proved by a two line computation, appears to be new. It provides a unified framework for controlling roots outside the substitution interval, replacing ad hoc convexity arguments that would need to be rederived for each degree.

3. **The boundary sign detection of exterior roots.** The observation that the signs of $f(0)$ and $f(\pi)$ detect exterior roots, combined with the parity argument that accounts for even- versus odd degree end behavior, provides a complete bridge between interior and exterior root counts.

4. **Uniformity across all degrees.** The framework applies identically to every degree $d \geq 4$, with only the number of residual cosine terms and the order of the exterior root derivative changing. Previous methods (discriminant, Sturm, Descartes) each have degree specific complications; the trigonometric method has a single template.

To summarize: the discriminant is exact but unwieldy; Descartes' rule is simple but inexact; Sturm's theorem is exact and general but computationally heavy. The trigonometric method occupies a different point in this landscape: it is conceptually light, geometrically transparent, and uniform across all degrees, at the cost of requiring $c_{d-2} < 0$ and a modest numerical analysis of the function f on a bounded interval.

9 Discussion

9.1 Why Chebyshev?

The Chebyshev polynomials T_d were introduced in the context of approximation theory: $T_d(x)/2^{d-1}$ is the monic polynomial of degree d with smallest maximum on $[-1, 1]$ [27]. The fact that they also provide a natural reduction of polynomial root classification to trigonometry is, to the best of the author's knowledge, not widely appreciated.

The connection is not accidental. The Chebyshev polynomial $T_d(x)$ has all d of its roots in $[-1, 1]$, equally spaced in the θ-domain (at $\theta = (2k-1)\pi/(2d)$). When we match a depressed polynomial to T_d via the cosine substitution, we are measuring how far the polynomial's root distribution deviates from the Chebyshev distribution. The residual function f encodes precisely this deviation. This echoes the philosophy of Erdős and Turán [10], who showed in their landmark *Annals of Mathematics* paper that the distribution of zeros of a polynomial on the unit circle is controlled by the size of its coefficients a qualitative connection between coefficient structure and root geometry that our method makes quantitatively explicit in a different setting.

9.2 The general pattern

The pattern across all degrees is now clear. For a degree d depressed polynomial with $c_{d-2} < 0$:

1. The Chebyshev identity absorbs the two highest surviving terms (t^d and $c_{d-2}t^{d-2}$).

2. The residual is a trigonometric function with $d-2$ free parameters.

3. The critical point equation is a polynomial of degree $d-1$ in $\cos\theta$.

4. The exterior root lemma ($P^{(d-2)} > 0$ outside $[-u, u]$) holds universally for all d.

5. The boundary signs of f detect exterior roots.

The complexity of the analysis grows mildly with d: the critical point equation increases by one degree for each step, and the residual gains one term. But the underlying mechanism bijection on $[0, \pi]$, boundary sign analysis, derivative cascade is identical for every degree.

9.3 Why not centuries ago?

The Chebyshev polynomials have been known since the mid-19th century, and the depressed form reduction of polynomials dates to the Renaissance. The substitution $t = u\cos\theta$ is a standard device in Vieta's trigonometric solution of the cubic. Why, then, wasn't this unified framework noticed long ago?

Perhaps the reason is that for the cubic, the trigonometric substitution was used to *solve* the equation (finding the roots explicitly), and the analogous goal for higher degrees is blocked by the Abel-Ruffini theorem [1] for $d \geq 5$. It may not have occurred to earlier mathematicians that the same substitution could be profitably used for the more modest but still useful goal of *classifying* the roots without finding them. The discriminant based classification was considered settled, and the 20th century's attention turned to other aspects of polynomial root geometry: the distribution of roots of random polynomials (Kac [16]), the Mahler measure and Lehmer's problem [3, 19], zeros of trigonometric polynomials [4], and the computational complexity of rootfinding (Smale [29], Basu-Pollack-Roy [2]). There was no obvious motivation to look for an alternative to the discriminant for the seemingly elementary problem of counting real roots. Rahman and Schmeisser [24] provide a comprehensive modern treatment of the analytic theory of polynomials that surveys many of these developments.

In the spirit of Loh [20], who showed that the quadratic formula admits a proof that might have been discovered four millennia earlier (see also earlier Monthly contributions by Heaton [12], Eells [9], and Cirul [6]), we hope that this trigonometric perspective may encourage the reader to look at familiar objects with fresh eyes. The interplay between polynomial equations and geometric or combinatorial structures has a long tradition in this journal (see, e.g., Sturmfels [31] and Savage [28]); the Chebyshev polynomials, which originated in approximation theory, turn out to be an elegant and uniform lens for viewing the geometry of polynomial roots at every degree.

References

[1] N. H. Abel, Mémoire sur les équations algébriques, où l'on démontre l'impossibilité de la résolution de l'équation générale du cinquième degré, *Œuvres Complètes*, vol. 1, 1824, pp. 28-33.

[2] S. Basu, R. Pollack, and M.-F. Roy, *Algorithms in Real Algebraic Geometry*, 2nd ed., Springer-Verlag, Berlin, 2006.

[3] P. Borwein, E. Dobrowolski, and M. J. Mossinghoff, Lehmer's problem for polynomials with odd coefficients, *Ann. of Math. (2)* **166** (2007), no. 2, 347-366.

[4] P. Borwein, T. Erdélyi, R. Ferguson, and R. Lockhart, On the zeros of cosine polynomials: solution to a problem of Littlewood, *Ann. of Math. (2)* **167** (2008), no. 3, 1109-1117.

[5] J. P. Boyd, Computing real roots of a polynomial in Chebyshev series form through subdivision, *Appl. Numer. Math.* **56** (2006), no. 8, 1077-1091.

[6] J. W. Cirul, A method for solving quadratic equations, *Amer. Math. Monthly* **44** (1937), no. 7, 462-463.

[7] R. Descartes, *La Géométrie*, Leiden, 1637.

[8] W. Dunham, Cardano and the solution of the cubic equation, Ch. 6 in *Journey through Genius: The Great Theorems of Mathematics*, Wiley, New York, 1990, pp. 133-154.

[9] W. C. Eells, Greek methods of solving quadratic equations, *Amer. Math. Monthly* **18** (1911), no. 1, 3-14.

[10] P. Erdős and P. Turán, On the distribution of roots of polynomials, *Ann. of Math. (2)* **51** (1950), no. 1, 105-119.

[11] I. M. Gelfand, M. M. Kapranov, and A. V. Zelevinsky, *Discriminants, Resultants, and Multidimensional Determinants*, Birkhäuser, Boston, 1994.

[12] H. Heaton, A method of solving quadratic equations, *Amer. Math. Monthly* **3** (1896), no. 10, 236-237.

[13] C. Hermite, Sur le nombre des racines d'une équation algébrique comprises entre des limites données, *J. Reine Angew. Math.* **52** (1856), 39-51.

[14] G. C. Holmes, The use of hyperbolic cosines in solving cubic polynomials, *Math. Gazette* **86** (2002), no. 507, 473-477.

[15] R. S. Irving, *Beyond the Quadratic Formula*, MAA, Washington, DC, 2013.

[16] M. Kac, On the average number of real roots of a random algebraic equation, *Bull. Amer. Math. Soc.* **49** (1943), 314-320.

[17] R. B. King, *Beyond the Quartic Equation*, Birkhäuser, Boston, 2009.

[18] D. Lazard, Quantifier elimination: optimal solution for two classical examples, *J. Symbolic Comput.* **32** (2001), 261-266.

[19] D. H. Lehmer, Factorization of certain cyclotomic functions, *Ann. of Math. (2)* **34** (1933), no. 3, 461-479.

[20] P.-S. Loh, A simple proof of the quadratic formula, *arXiv:1910.06709*, 2019.

[21] J. C. Mason and D. C. Handscomb, *Chebyshev Polynomials*, Chapman & Hall/CRC, Boca Raton, FL, 2003.

[22] M. Marden, *Geometry of Polynomials*, 2nd ed., Mathematical Surveys and Monographs, vol. 3, AMS, Providence, RI, 1966.

[23] R. W. D. Nickalls, Vieta, Descartes and the cubic equation, *Math. Gazette* **90** (2006), no. 518, 203-208.

[24] Q. I. Rahman and G. Schmeisser, *Analytic Theory of Polynomials*, Oxford University Press, 2002.

[25] P. K. Rees, The discriminant of a quartic equation, *Amer. Math. Monthly* **29** (1922), no. 4, 150-153.

[26] P. K. Rees, Graphical discussion of the roots of a quartic equation, *Amer. Math. Monthly* **29** (1922), no. 2, 51-55.

[27] T. J. Rivlin, *Chebyshev Polynomials: From Approximation Theory to Algebra and Number Theory*, 2nd ed., Wiley, New York, 1990.

[28] J. Savage, Factoring quadratics, *Math. Teacher* **82** (1989), no. 1, 35-36.

[29] S. Smale, The fundamental theorem of algebra and complexity theory, *Bull. Amer. Math. Soc. (N.S.)* **4** (1981), no. 1, 1-36.

[30] C. Sturm, Mémoire sur la résolution des équations numériques, *Bull. Sci. Férussac* **11** (1829), 419-425.

[31] B. Sturmfels, Polynomial equations and convex polytopes, *Amer. Math. Monthly* **105** (1998), no. 10, 907-922.

[32] J. J. Sylvester, On a theory of the syzygetic relations of two rational integral functions, *Philos. Trans. Roy. Soc. London* **143** (1853), 407-548.

[33] L. N. Trefethen, *Spectral Methods in MATLAB*, SIAM, Philadelphia, 2000.

[34] S. Pratiher, A simple trigonometric classification of quartic roots, *arXiv:2603.28814*, 2026.

[35] S. Pratiher, A simple trigonometric classification of quintic roots, *arXiv:2603.28352*, 2026.

[36] F. Viète, *Opera Mathematica*, Leiden, 1646. (For a modern treatment, see D. Cox, J. Little, and D. O'Shea, *Ideals, Varieties, and Algorithms*, 4th ed., Springer, 2015.)